CS 418 | Final Project Red Wine Quality Analysis

Predicting Red Wine Quality:

Physicochemical Approach using Machine Learning

Team Information

- Team Number: #15

- Team Members:

Name	NetID	UIN
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Instructions for Running Code

(more information in README file)

1. Open the project notebook in **Google Colab**
2. Click the **"Run all"** button on the top toolbar
3. Select **"Choose Files"** button on the left side of the Upload cell (waiting for input data)
4. From your local computer, upload the following **CSV file**:
   - `winequality-red.csv`
5. After uploading, all remaining cells should execute properly.

In [39]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Upload dataset: 'winequality-red.csv'
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving winequality-red.csv to winequality-red (1).csv
```

In [40]:
```
sns.set_theme(font="Serif")
```

Data Cleaning & Processing (Brian)

In [41]:
```
# 1. Load & inspect dataset
df = pd.read_csv('winequality-red.csv', sep = ';')

print("Load & Inspect")
print(f"Shape: {df.shape}\n")
df.info()
display(df.describe())
```

```
Load & Inspect
Shape: (1599, 12)

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1599 entries, 0 to 1598
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   fixed acidity         1599 non-null   float64
 1   volatile acidity      1599 non-null   float64
 2   citric acid           1599 non-null   float64
 3   residual sugar        1599 non-null   float64
 4   chlorides             1599 non-null   float64
 5   free sulfur dioxide   1599 non-null   float64
 6   total sulfur dioxide  1599 non-null   float64
 7   density               1599 non-null   float64
 8   pH                    1599 non-null   float64
 9   sulphates             1599 non-null   float64
 10  alcohol               1599 non-null   float64
 11  quality               1599 non-null   int64
dtypes: float64(11), int64(1)
memory usage: 150.0 KB
```

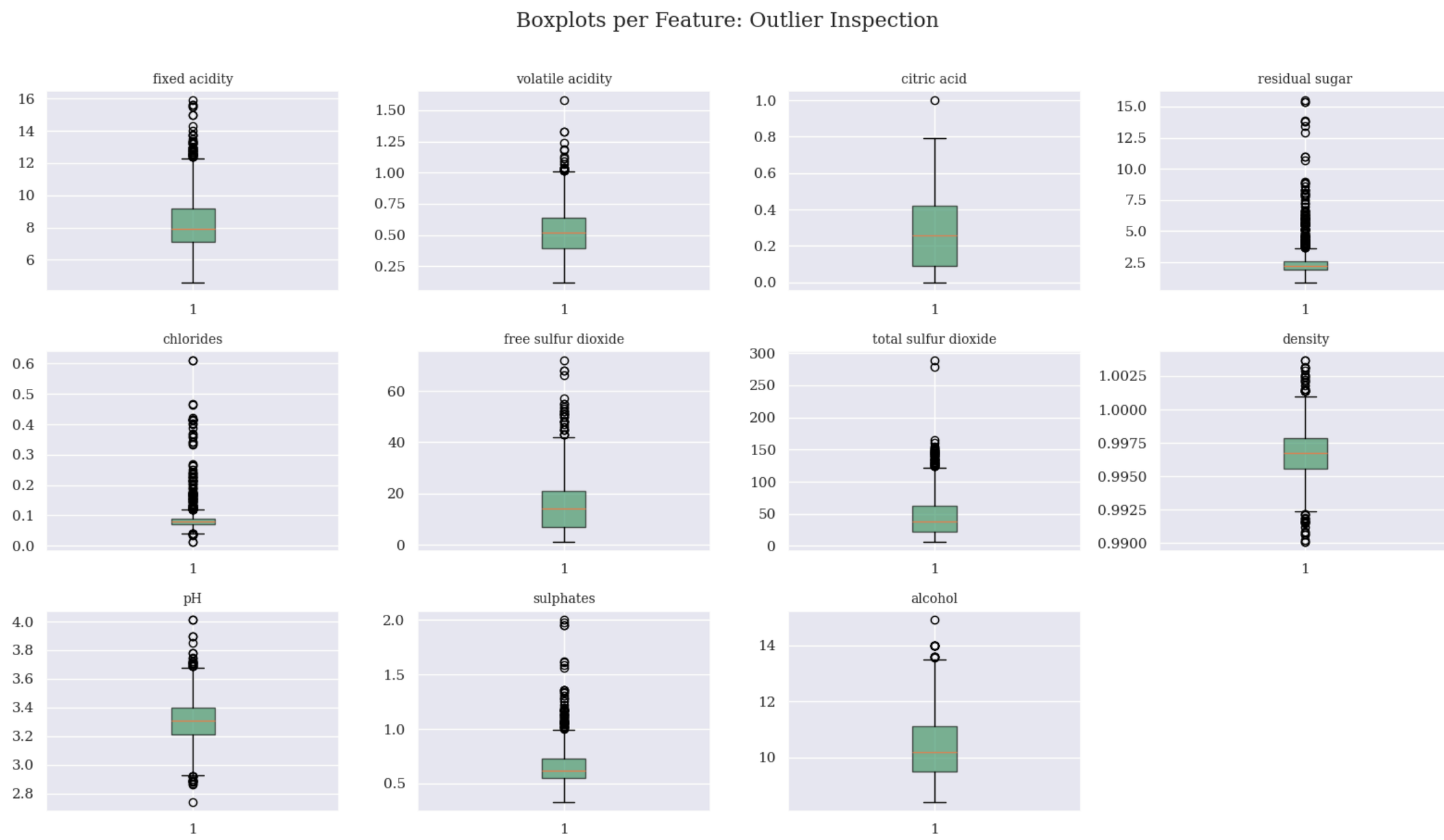
	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
count	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000
mean	8.319637	0.527821	0.270976	2.538806	0.087467	15.874922	46.467792	0.996747	3.311113	0.658149	10.422983	5.636023
std	1.741096	0.179060	0.194801	1.409928	0.047065	10.460157	32.895324	0.001887	0.154386	0.169507	1.065668	0.807569
min	4.600000	0.120000	0.000000	0.900000	0.012000	1.000000	6.000000	0.990070	2.740000	0.330000	8.400000	3.000000
25%	7.100000	0.390000	0.090000	1.900000	0.070000	7.000000	22.000000	0.995600	3.210000	0.550000	9.500000	5.000000
50%	7.900000	0.520000	0.260000	2.200000	0.079000	14.000000	38.000000	0.996750	3.310000	0.620000	10.200000	6.000000
75%	9.200000	0.640000	0.420000	2.600000	0.090000	21.000000	62.000000	0.997835	3.400000	0.730000	11.100000	6.000000
max	15.900000	1.580000	1.000000	15.500000	0.611000	72.000000	289.000000	1.003690	4.010000	2.000000	14.900000	8.000000

In [42]:
```
# 2. Missing value check
print("\nMissing Value Check")
print("\n...")
missing = df.isnull().sum()
if missing.any():
    print(missing[missing > 0])
else:
    print("Dataset is clean!")
```

```
Missing Value Check

...
Dataset is clean!
```

In [43]:
```
# 3. Outlier detection
feature_cols = [c for c in df.columns if (c != 'quality' and c != 'quality_label' and c != 'quality_label_name')]

# Convert columns to numeric if needed
for col in feature_cols:
```

```
    df[col] = pd.to_numeric(df[col], errors='coerce')

# Boxplots for visual inspection
n_cols = 4
n_rows = int(np.ceil(len(feature_cols) / n_cols))

fig, axes = plt.subplots(n_rows, n_cols, figsize=(16, n_rows * 3))
axes = axes.flatten()

for i, col in enumerate(feature_cols):
    axes[i].boxplot(df[col].dropna(), patch_artist=True,
                    boxprops=dict(facecolor='seagreen', alpha=0.6))
    axes[i].set_title(col, fontsize=10)
    axes[i].set_xlabel('')

for j in range(i + 1, len(axes)):
    axes[j].set_visible(False)

plt.suptitle('Boxplots per Feature: Outlier Inspection', fontsize=16, y=1.01)
plt.tight_layout()
plt.show()
```



In [44]:
```
print("\nIQR Outlier Flag Summary (1.5×IQR rule)")
for col in feature_cols:
    Q1  = df[col].quantile(0.25)
    Q3  = df[col].quantile(0.75)
    IQR = Q3 - Q1
    n_out = ((df[col] < Q1 - 1.5 * IQR) | (df[col] > Q3 + 1.5 * IQR)).sum()
    print(f"  {col:30s}: {n_out:3d} flagged outliers")
print("\nRows are NOT dropped!")
```

```
IQR Outlier Flag Summary (1.5×IQR rule)
  fixed acidity                 :  49 flagged outliers
  volatile acidity              :  19 flagged outliers
  citric acid                   :   1 flagged outliers
  residual sugar                : 155 flagged outliers
  chlorides                     : 112 flagged outliers
  free sulfur dioxide           :  30 flagged outliers
  total sulfur dioxide          :  55 flagged outliers
  density                       :  45 flagged outliers
  pH                            :  35 flagged outliers
  sulphates                     :  59 flagged outliers
  alcohol                       :  13 flagged outliers

Rows are NOT dropped!
```

In [45]:
```
# 4. Feature Engineering
df['quality_label'] = (df['quality'] >= 7).astype(int)

print("\nquality_label Distribution")
print("\n...")

vc = df['quality_label'].value_counts().rename(index={1: 'Good (>=7)', 0: 'Not Good (<7)'})
print(vc)

print(f"\nGood wine rate: {df['quality_label'].mean():.1%}")
```

```
quality_label Distribution

...
quality_label
Not Good (<7)    1382
Good (>=7)        217
Name: count, dtype: int64

Good wine rate: 13.6%
```

In [46]:
```
#5. Train/Test split
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X_all = df[feature_cols]  # all physicochemical features
y_reg = df['quality']  # regression target
y_clf = df['quality_label']  # classification target (binary)

X_train, X_test, y_train_reg, y_test_reg, y_train_clf, y_test_clf = train_test_split(
    X_all, y_reg, y_clf,
    test_size=0.20,
    random_state=42,
    stratify=y_clf
```

```
)

scaler = StandardScaler()
X_train_sc = pd.DataFrame(scaler.fit_transform(X_train), columns=feature_cols)
X_test_sc  = pd.DataFrame(scaler.transform(X_test), columns=feature_cols)

print("\nTrain/Test Split")
print("...")
print(f"Training set : {X_train_sc.shape} | Good rate: {y_train_clf.mean():.1%}")
print(f"Test set     : {X_test_sc.shape}  | Good rate: {y_test_clf.mean():.1%}")
print("\nData Preparation complete!")
```

```
Train/Test Split
...
Training set : (1279, 11) | Good rate: 13.6%
Test set     : (320, 11)  | Good rate: 13.4%

Data Preparation complete!
```

Exploratory Data Analysis (Han)

- Summary Stats (done above)
- Correlation heatmap
- Feature Importance (correlation)
- Histogram & bar chart: Quality Score Distribution
- Scatterplot: two high-impact features (alcohol & volatile acidity)
- Box plots: All features by Quality Label(good/not) & Quality Scores(3-8)

In [47]:
```
# build own sns palette
sns.color_palette('Set2',2)
```

Out[47]:

In [48]:
```
color = sns.color_palette('Set2',2)
sns.set_palette(color)
print(color)
```

```
[(0.4, 0.7607843137254902, 0.6470588235294118), (0.9882352941176471, 0.5529411764705883, 0.3843137254901961)]
```

In [49]:
```
# add new column with classification label
df['quality_label_name'] = df['quality_label'].map({0: 'Not Good (<7)', 1: 'Good (>=7)'})
```

In [50]:
```
# summary statistics (+1 columns)
df.info()
df.describe()
```

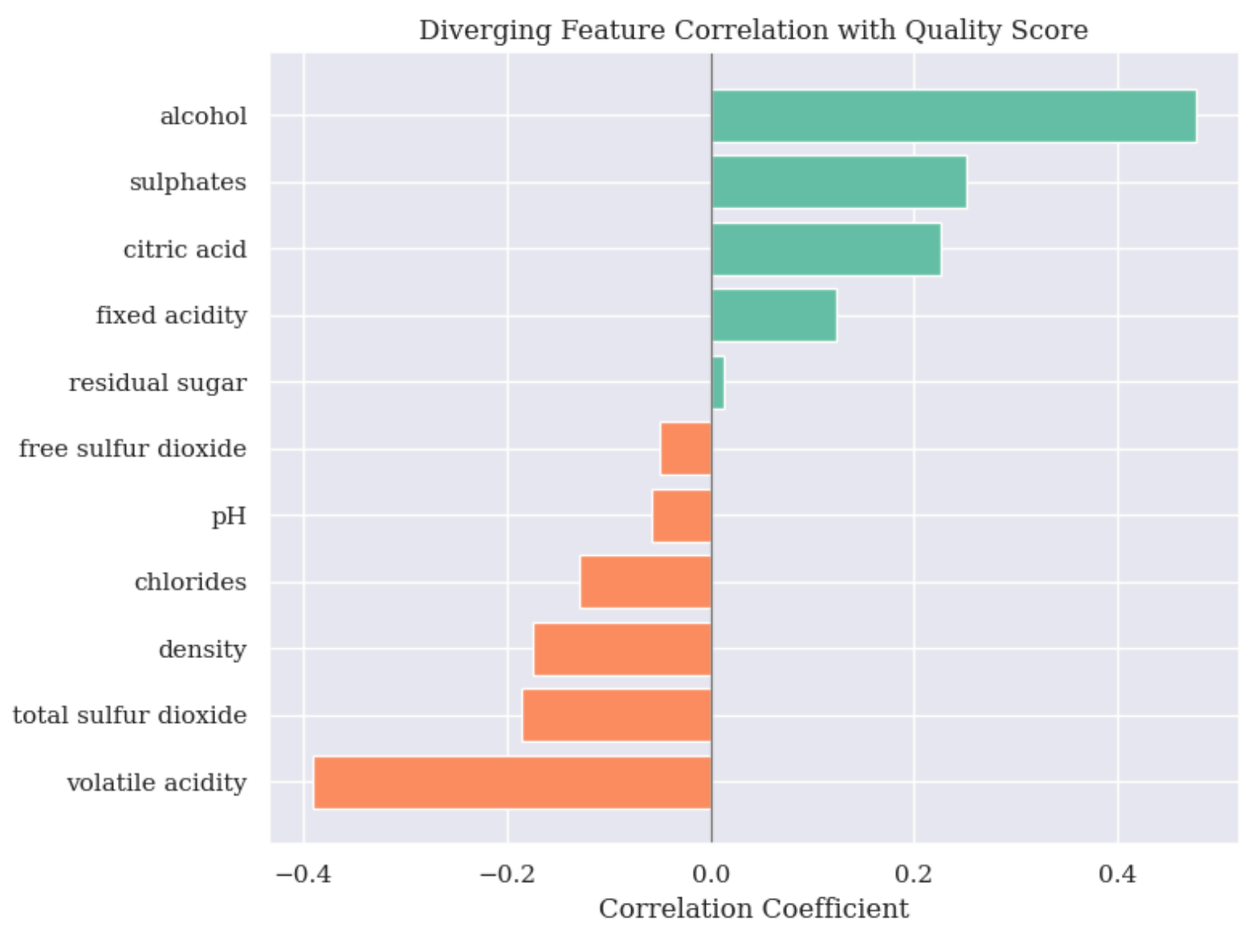
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1599 entries, 0 to 1598
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   fixed acidity         1599 non-null   float64
 1   volatile acidity      1599 non-null   float64
 2   citric acid           1599 non-null   float64
 3   residual sugar        1599 non-null   float64
 4   chlorides             1599 non-null   float64
 5   free sulfur dioxide   1599 non-null   float64
 6   total sulfur dioxide  1599 non-null   float64
 7   density               1599 non-null   float64
 8   pH                    1599 non-null   float64
 9   sulphates             1599 non-null   float64
 10  alcohol               1599 non-null   float64
 11  quality               1599 non-null   int64
 12  quality_label         1599 non-null   int64
 13  quality_label_name    1599 non-null   object
dtypes: float64(11), int64(2), object(1)
memory usage: 175.0+ KB
```

Out[50]:

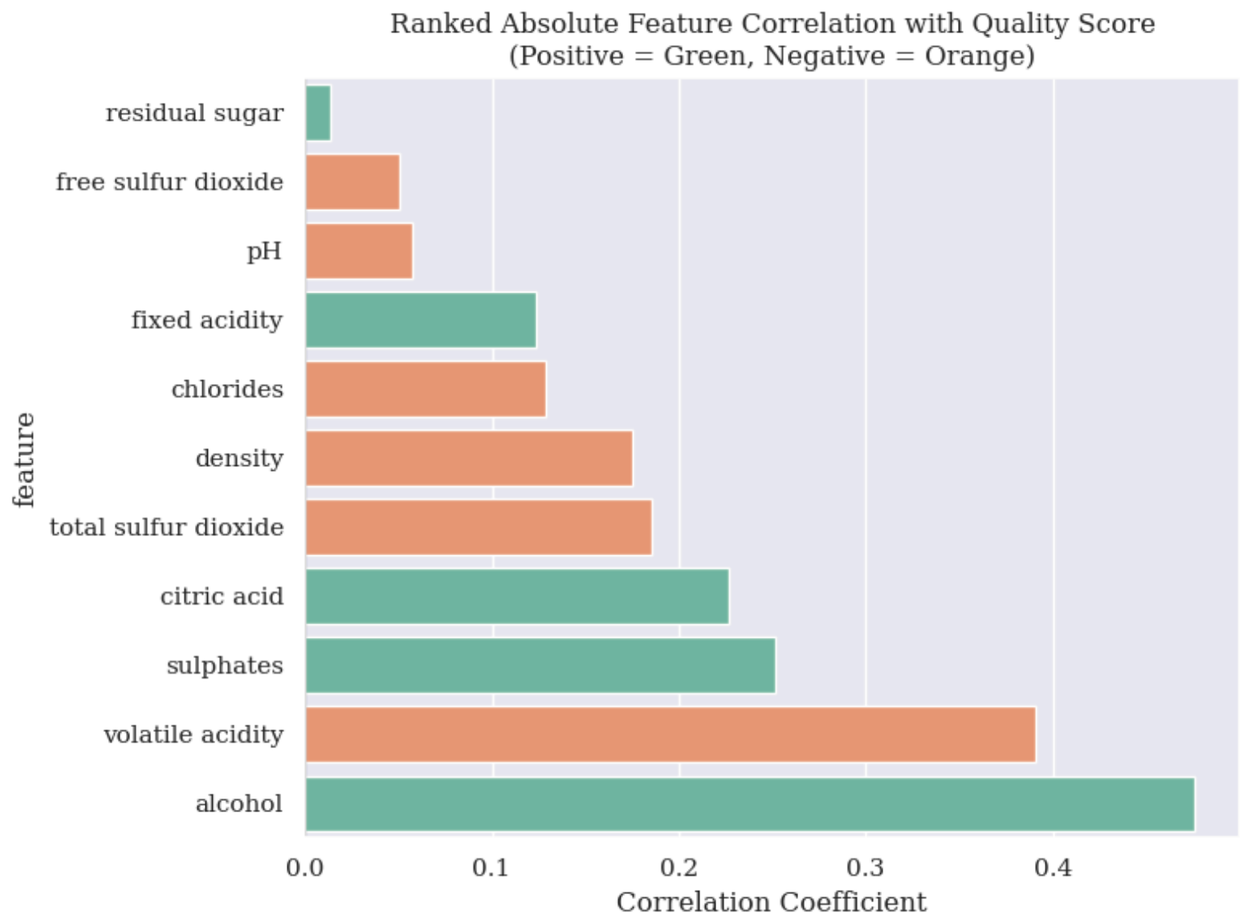
	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality	quality_label
count	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000	1599.000000
mean	8.319637	0.527821	0.270976	2.538806	0.087467	15.874922	46.467792	0.996747	3.311113	0.658149	10.422983	5.636023	0.135710
std	1.741096	0.179060	0.194801	1.409928	0.047065	10.460157	32.895324	0.001887	0.154386	0.169507	1.065668	0.807569	0.342587
min	4.600000	0.120000	0.000000	0.900000	0.012000	1.000000	6.000000	0.990070	2.740000	0.330000	8.400000	3.000000	0.000000
25%	7.100000	0.390000	0.090000	1.900000	0.070000	7.000000	22.000000	0.995600	3.210000	0.550000	9.500000	5.000000	0.000000
50%	7.900000	0.520000	0.260000	2.200000	0.079000	14.000000	38.000000	0.996750	3.310000	0.620000	10.200000	6.000000	0.000000
75%	9.200000	0.640000	0.420000	2.600000	0.090000	21.000000	62.000000	0.997835	3.400000	0.730000	11.100000	6.000000	0.000000
max	15.900000	1.580000	1.000000	15.500000	0.611000	72.000000	289.000000	1.003690	4.010000	2.000000	14.900000	8.000000	1.000000

In [51]:
```
# Correlation heatmap
from matplotlib.colors import LinearSegmentedColormap
custom_cmap = LinearSegmentedColormap.from_list(
    'green_orange',
     #['coral', 'white', 'mediumseagreen']
    [(0.9882352941176471, 0.5529411764705883, 0.3843137254901961),
     'white',(0.4, 0.7607843137254902, 0.6470588235294118) ]
)
plt.figure(figsize=(12, 9))
corr_matrix = df.drop(columns=['quality_label', 'quality_label_name']).corr()
sns.heatmap(corr_matrix, annot=True, fmt=".2f",
            cmap=custom_cmap, #plt.cm.Blues or RdYlGn
            square=True, linewidths=0.5)
plt.title('Correlation Heatmap: All Features + Quality', fontsize=16)
plt.tight_layout()
plt.show()
```

In [52]:
```
# get correlations with quality, drop non-numb(quality, quality_label and quality_label_name) themselves
quality_corr = df.select_dtypes(include='number').corr()['quality'].drop(['quality', 'quality_label']).sort_values()
# set color: positive = green, negative = orange
colors = [color[1] if v < 0 else color[0] for v in quality_corr.values]
plt.figure(figsize=(8, 6))
plt.barh(quality_corr.index, quality_corr.values, color=colors)
plt.axvline(x=0, color='gray', linewidth=1)  #centre line
plt.title('Diverging Feature Correlation with Quality Score')
plt.xlabel('Correlation Coefficient')
plt.tight_layout()
plt.show()
```

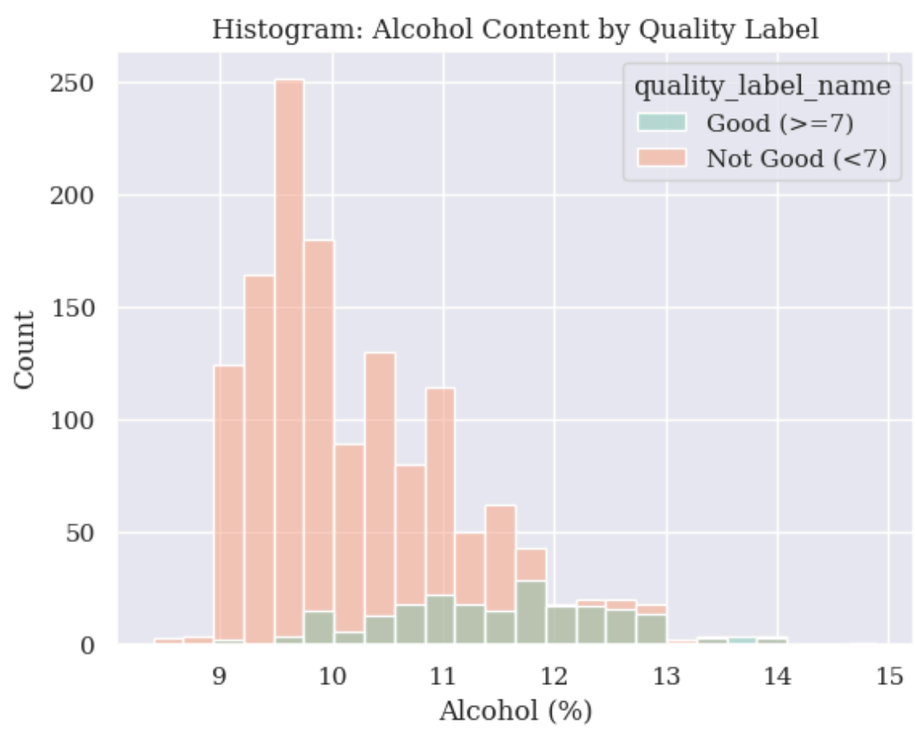


In [53]:
```
# feature importance, ranked high-low (absolute correlation with quality)
rank_df = (df.select_dtypes(include='number').corr()['quality'].drop(['quality', 'quality_label']).sort_values().reset_index())
rank_df.columns = ['feature', 'correlation']
rank_df['direction'] = rank_df['correlation'].apply(lambda x: 'negative' if x < 0 else 'positive')
rank_df['abs_correlation'] = rank_df['correlation'].abs()
rank_df = rank_df.sort_values('abs_correlation')
plt.figure(figsize=(8, 6))
sns.barplot(
    data=rank_df,
    x='abs_correlation',
    y='feature',
    hue='direction',
    palette={'positive': color[0], 'negative': color[1]},
    legend=False
)
plt.axvline(x=0, color='black', linewidth=0.8)
plt.title('Ranked Absolute Feature Correlation with Quality Score\n(Positive = Green, Negative = Orange)')
plt.xlabel('Correlation Coefficient')
plt.tight_layout()
plt.show()
```



Distribution of quality scores (histogram + bar chart)

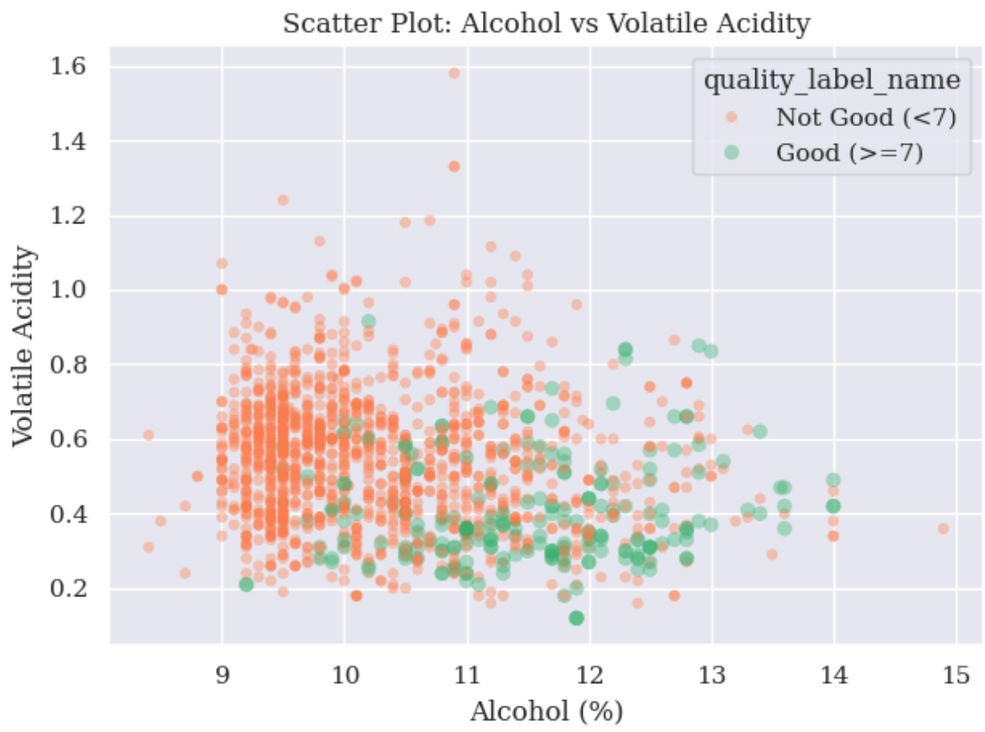
In [54]:
```
# 1. Histogram: alchol dist by quality label
sns.histplot(
    data=df,
    x='alcohol',
    hue='quality_label_name',
    hue_order=['Good (>=7)', 'Not Good (<7)'],
    alpha=0.4,
    palette={'Good (>=7)': color[0], 'Not Good (<7)': color[1]} #'hls' #ref:husl, paired, pastel, dark
)
plt.title('Histogram: Alcohol Content by Quality Label')
plt.xlabel('Alcohol (%)')
plt.ylabel('Count')
plt.show()
```



In [55]:
```
# 2. Bar chart: Wine Count per Quality Score
my_palette = sns.light_palette('seagreen', n_colors=9)
my_palette.reverse()
sns.countplot(
    data=df,
    x='quality',
    palette=my_palette)
plt.title('Bar Chart: Wine Count per Quality Score')
plt.xlabel('Quality Score')
plt.ylabel('Wine Count')
plt.show()
```

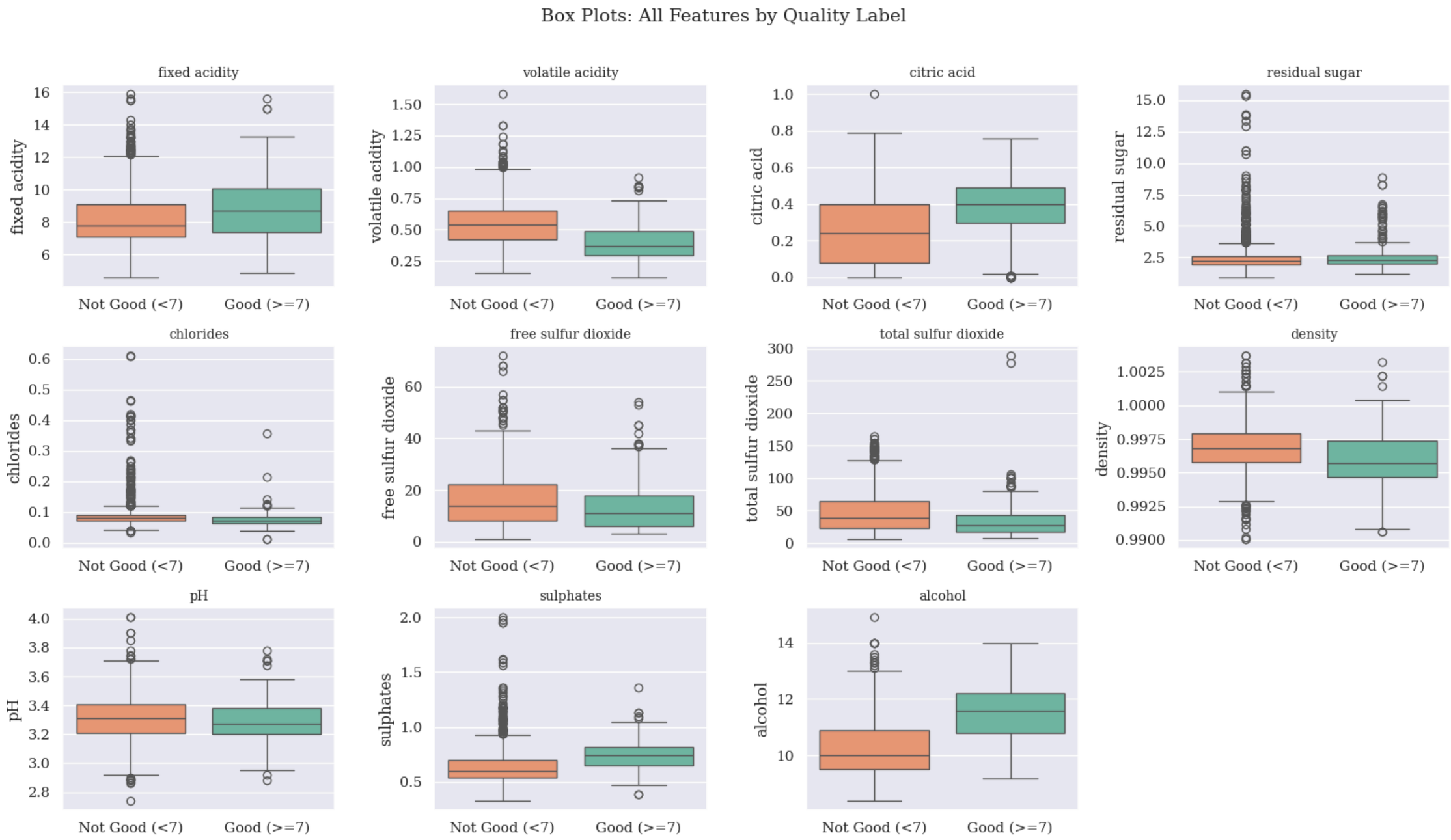
```
/tmp/ipykernel_1375/713337836.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(
/tmp/ipykernel_1375/713337836.py:4: UserWarning: The palette list has more values (9) than needed (6), which may not be intended.
  sns.countplot(
```

In [58]:
```
# 3. Scatterplot: alcohol vs volatile acidity, colored by quality label
sns.scatterplot(data=df, x='alcohol', y='volatile acidity', hue='quality_label_name',
                palette={'Good (>=7)': 'mediumseagreen', 'Not Good (<7)': 'coral'},
                #  {'Good (>=7)': color[0], 'Not Good (<7)': color[1]}, #Set2
                alpha=0.4, # self-note: for overlapping transparency
                edgecolor='None', # for no dot border color
                size='quality_label_name',
                sizes={'Good (>=7)': 45, 'Not Good (<7)': 25}
)
plt.title('Scatter Plot: Alcohol vs Volatile Acidity')
plt.xlabel('Alcohol (%)')
plt.ylabel('Volatile Acidity')
plt.tight_layout()
plt.show()
```



In [56]:
```
# 4. Box plots: all features by quality label, scores
fig, axes = plt.subplots(n_rows, n_cols, figsize=(16, n_rows * 3))
axes = axes.flatten()

for i, col in enumerate(feature_cols):
    sns.boxplot(
        data=df,
        x='quality_label_name',
        y=col,
        palette={'Good (>=7)': color[0], 'Not Good (<7)': color[1]},
        ax=axes[i]
    )
    axes[i].set_title(col, fontsize=10)
    axes[i].set_xlabel('')  # cleaner, removes repeated x label

for j in range(i + 1, len(axes)):
    axes[j].set_visible(False)

plt.suptitle('Box Plots: All Features by Quality Label', fontsize=14, y=1.01)
plt.tight_layout()
plt.show()
```

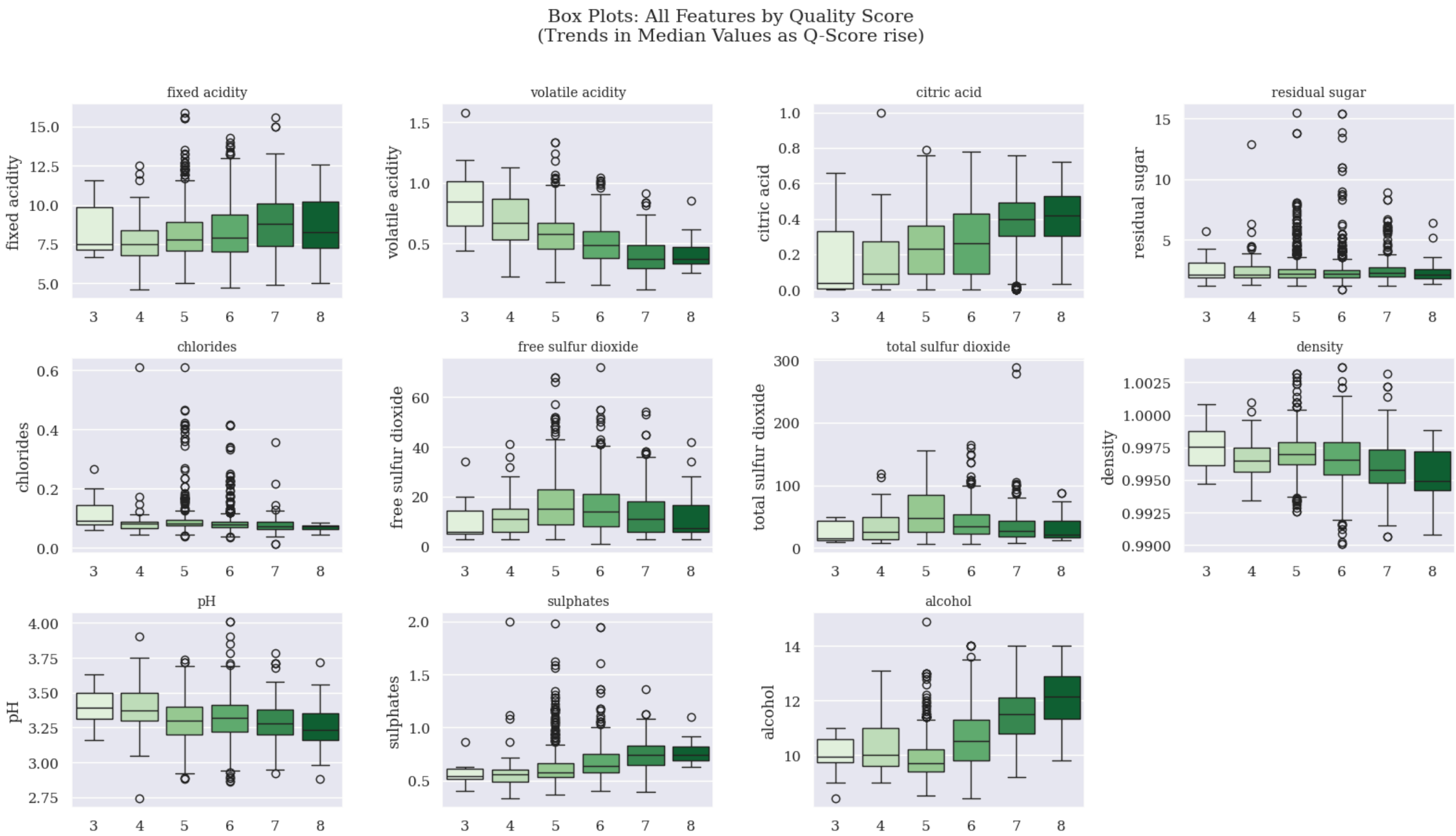
```
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/363722531.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
```



In [57]:

```
fig, axes = plt.subplots(n_rows, n_cols, figsize=(16, n_rows * 3))
axes = axes.flatten()

for i, col in enumerate(feature_cols):
    sns.boxplot(
        data=df,
        x='quality',
        y=col,
        palette='Greens',
        ax=axes[i]
    )
    axes[i].set_title(col, fontsize=10)
    axes[i].set_xlabel('')  # cleaner, removes repeated x label

for j in range(i + 1, len(axes)):
    axes[j].set_visible(False)

plt.suptitle('Box Plots: All Features by Quality Score\n'+
'(Trends in Median Values as Q-Score rise)', fontsize=14, y=1.01)
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
/tmp/ipykernel_1375/2904092073.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
```



Regression (Brian)

5 LR models

In [59]:
```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Feature combo definitions

COMBOS = {
    1: {
        'name': 'All features ',
        'features': feature_cols,
    },
    2: {
        'name': 'Top 3 known predictors',
        'features': ['alcohol', 'volatile acidity', 'sulphates'],
    },
    3: {
        'name' : 'Flavor profile',
        'features': ['alcohol', 'volatile acidity', 'citric acid', 'density'],
    },
    4: {
        'name' : 'Top 5 by correlation',
        'features': ['alcohol', 'volatile acidity', 'sulphates',
                     'citric acid', 'total sulfur dioxide'],
    },
    5: {
        'name' : 'Low-collinearity set',
        'features': ['alcohol', 'volatile acidity', 'sulphates',
                     'pH', 'residual sugar'],
    },
}

# Run all 5 combos; same scaler fitted on X_train above
regression_results = []

for combo_id, combo in COMBOS.items():
    feats = combo['features']
    model = LinearRegression()
    model.fit(X_train_sc[feats], y_train_reg)

    y_pred_train = model.predict(X_train_sc[feats])
    y_pred_test  = model.predict(X_test_sc[feats])

    train_rmse = np.sqrt(mean_squared_error(y_train_reg, y_pred_train))
    test_rmse  = np.sqrt(mean_squared_error(y_test_reg,  y_pred_test))
    train_r2   = r2_score(y_train_reg, y_pred_train)
    test_r2    = r2_score(y_test_reg,  y_pred_test)

    regression_results.append({
        'Combo': combo_id,
        'Name' : combo['name'],
        'Features Used': ', '.join(feats),
        'Num Features': len(feats),
        'Train RMSE' : round(train_rmse, 4),
        'Test RMSE': round(test_rmse,  4),
        'Train R2' : round(train_r2,   4),
        'Test R2': round(test_r2,    4),
    })

    print(f"\nCombo {combo_id}: {combo['name']}")
    print("...")
    print(f"Features : {feats}")
    print(f"Train RMSE : {train_rmse:.4f}  |  Test RMSE : {test_rmse:.4f}")
    print(f"Train R2: {train_r2:.4f}     |  Test R2   : {test_r2:.4f}")
```

```
Combo 1: All features
...
Features : ['fixed acidity', 'volatile acidity', 'citric acid', 'residual sugar', 'chlorides', 'free sulfur dioxide', 'total sulfur dioxide', 'density', 'pH', 'sulphates', 'alco
hol']
Train RMSE : 0.6409  |  Test RMSE : 0.6687
Train R2: 0.3484     |  Test R2   : 0.3916

Combo 2: Top 3 known predictors
...
Features : ['alcohol', 'volatile acidity', 'sulphates']
Train RMSE : 0.6557  |  Test RMSE : 0.6694
Train R2: 0.3181     |  Test R2   : 0.3903

Combo 3: Flavor profile
...
Features : ['alcohol', 'volatile acidity', 'citric acid', 'density']
Train RMSE : 0.6610  |  Test RMSE : 0.6896
Train R2: 0.3069     |  Test R2   : 0.3530

Combo 4: Top 5 by correlation
...
Features : ['alcohol', 'volatile acidity', 'sulphates', 'citric acid', 'total sulfur dioxide']
Train RMSE : 0.6497  |  Test RMSE : 0.6756
Train R2: 0.3305     |  Test R2   : 0.3790

Combo 5: Low-collinearity set
...
Features : ['alcohol', 'volatile acidity', 'sulphates', 'pH', 'residual sugar']
Train RMSE : 0.6542  |  Test RMSE : 0.6683
Train R2: 0.3211     |  Test R2   : 0.3922
```

Classification (Mahsa)

Decision Tree, KNN, Naive Bayes, Logistic Regression (bonus)

In [60]:
```
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, classification_report

# Feature combinations for classification
classification_combos = {
    "All 11 features": feature_cols,
    "Top 5 by correlation": ['alcohol', 'volatile acidity', 'sulphates',
                             'citric acid', 'total sulfur dioxide']
}

print("Classification setup complete.")
print("Target variable: quality_label")
print("Feature combos:")
for name, feats in classification_combos.items():
    print(name, ":", feats)
```

```
Classification setup complete.
Target variable: quality_label
Feature combos:
All 11 features : ['fixed acidity', 'volatile acidity', 'citric acid', 'residual sugar', 'chlorides', 'free sulfur dioxide', 'total sulfur dioxide', 'density', 'pH', 'sulphate
s', 'alcohol']
Top 5 by correlation : ['alcohol', 'volatile acidity', 'sulphates', 'citric acid', 'total sulfur dioxide']
```

In [61]:
```
# Create scaled DataFrames for classification
X_train_scaled_df = pd.DataFrame(X_train, columns=feature_cols)
X_test_scaled_df = pd.DataFrame(X_test, columns=feature_cols)

y_train_class = y_train_clf
y_test_class = y_test_clf

print(X_train_scaled_df.shape)
print(X_test_scaled_df.shape)
```

```
(1279, 11)
(320, 11)
```

Decision Tree Classification

In [62]:
```
# Decision Tree Classification
dt_results = []

for combo_name, features in classification_combos.items():
    X_train_combo = X_train_scaled_df[features]
    X_test_combo = X_test_scaled_df[features]

    for depth in [3, 5, 10, None]:
        for min_split in [2, 5, 10]:
            dt_model = DecisionTreeClassifier(
                criterion='entropy',
                max_depth=depth,
                min_samples_split=min_split,
                random_state=42
            )

            dt_model.fit(X_train_combo, y_train_class)
            y_pred = dt_model.predict(X_test_combo)

            dt_results.append({
                "Model": "Decision Tree",
                "Features": combo_name,
                "Max Depth": depth,
                "Min Samples Split": min_split,
                "Accuracy": accuracy_score(y_test_class, y_pred),
                "Precision": precision_score(y_test_class, y_pred, zero_division=0),
                "Recall": recall_score(y_test_class, y_pred, zero_division=0),
                "F1 Score": f1_score(y_test_class, y_pred, zero_division=0)
            })

dt_results_df = pd.DataFrame(dt_results)
dt_results_df.sort_values(by="F1 Score", ascending=False)
```

Out[62]:

	Model	Features	Max Depth	Min Samples Split	Accuracy	Precision	Recall	F1 Score
10	Decision Tree	All 11 features	NaN	5	0.925000	0.731707	0.697674	0.714286
9	Decision Tree	All 11 features	NaN	2	0.925000	0.731707	0.697674	0.714286
11	Decision Tree	All 11 features	NaN	10	0.918750	0.717949	0.651163	0.682927
6	Decision Tree	All 11 features	10.0	2	0.912500	0.692308	0.627907	0.658537
7	Decision Tree	All 11 features	10.0	5	0.909375	0.694444	0.581395	0.632911
8	Decision Tree	All 11 features	10.0	10	0.906250	0.675676	0.581395	0.625000
4	Decision Tree	All 11 features	5.0	5	0.906250	0.709677	0.511628	0.594595
5	Decision Tree	All 11 features	5.0	10	0.906250	0.709677	0.511628	0.594595
3	Decision Tree	All 11 features	5.0	2	0.903125	0.687500	0.511628	0.586667
22	Decision Tree	Top 5 by correlation	NaN	5	0.884375	0.568182	0.581395	0.574713
18	Decision Tree	Top 5 by correlation	10.0	2	0.884375	0.571429	0.558140	0.564706
20	Decision Tree	Top 5 by correlation	10.0	10	0.890625	0.611111	0.511628	0.556962
21	Decision Tree	Top 5 by correlation	NaN	2	0.884375	0.575000	0.534884	0.554217
16	Decision Tree	Top 5 by correlation	5.0	5	0.903125	0.730769	0.441860	0.550725
17	Decision Tree	Top 5 by correlation	5.0	10	0.903125	0.730769	0.441860	0.550725
19	Decision Tree	Top 5 by correlation	10.0	5	0.881250	0.560976	0.534884	0.547619
15	Decision Tree	Top 5 by correlation	5.0	2	0.900000	0.703704	0.441860	0.542857
23	Decision Tree	Top 5 by correlation	NaN	10	0.887500	0.600000	0.488372	0.538462
2	Decision Tree	All 11 features	3.0	10	0.893750	0.714286	0.348837	0.468750
0	Decision Tree	All 11 features	3.0	2	0.893750	0.714286	0.348837	0.468750
1	Decision Tree	All 11 features	3.0	5	0.893750	0.714286	0.348837	0.468750
14	Decision Tree	Top 5 by correlation	3.0	10	0.893750	0.714286	0.348837	0.468750
12	Decision Tree	Top 5 by correlation	3.0	2	0.893750	0.714286	0.348837	0.468750
13	Decision Tree	Top 5 by correlation	3.0	5	0.893750	0.714286	0.348837	0.468750

KNN Classification

In [63]:

```
# KNN Classification

knn_results = []

for combo_name, features in classification_combos.items():
    X_train_combo = X_train_scaled_df[features]
    X_test_combo = X_test_scaled_df[features]

    for k in [3, 5, 7, 11]:
        knn_model = KNeighborsClassifier(n_neighbors=k)

        knn_model.fit(X_train_combo, y_train_class)
        y_pred = knn_model.predict(X_test_combo)

        knn_results.append({
            "Model": "KNN",
            "Features": combo_name,
            "K": k,
            "Accuracy": accuracy_score(y_test_class, y_pred),
            "Precision": precision_score(y_test_class, y_pred, zero_division=0),
            "Recall": recall_score(y_test_class, y_pred, zero_division=0),
            "F1 Score": f1_score(y_test_class, y_pred, zero_division=0)
        })

knn_results_df = pd.DataFrame(knn_results)
knn_results_df.sort_values(by="F1 Score", ascending=False)
```

Out[63]:

	Model	Features	K	Accuracy	Precision	Recall	F1 Score
4	KNN	Top 5 by correlation	3	0.871875	0.531250	0.395349	0.453333
5	KNN	Top 5 by correlation	5	0.865625	0.500000	0.325581	0.394366
0	KNN	All 11 features	3	0.840625	0.388889	0.325581	0.354430
2	KNN	All 11 features	7	0.871875	0.550000	0.255814	0.349206
6	KNN	Top 5 by correlation	7	0.868750	0.526316	0.232558	0.322581
1	KNN	All 11 features	5	0.859375	0.454545	0.232558	0.307692
7	KNN	Top 5 by correlation	11	0.871875	0.600000	0.139535	0.226415
3	KNN	All 11 features	11	0.859375	0.400000	0.093023	0.150943

Naive Bayes Classification

In [64]:

```
from sklearn.naive_bayes import GaussianNB

# Naive Bayes Classification

nb_results = []

for combo_name, features in classification_combos.items():
    X_train_combo = X_train_scaled_df[features]
    X_test_combo = X_test_scaled_df[features]

    nb_model = GaussianNB()

    nb_model.fit(X_train_combo, y_train_class)
    y_pred = nb_model.predict(X_test_combo)

    nb_results.append({
        "Model": "Naive Bayes",
        "Features": combo_name,
        "Accuracy": accuracy_score(y_test_class, y_pred),
        "Precision": precision_score(y_test_class, y_pred, zero_division=0),
        "Recall": recall_score(y_test_class, y_pred, zero_division=0),
        "F1 Score": f1_score(y_test_class, y_pred, zero_division=0)
    })

nb_results_df = pd.DataFrame(nb_results)
nb_results_df.sort_values(by="F1 Score", ascending=False)
```

Out[64]:

	Model	Features	Accuracy	Precision	Recall	F1 Score
0	Naive Bayes	All 11 features	0.865625	0.500000	0.720930	0.590476
1	Naive Bayes	Top 5 by correlation	0.884375	0.568182	0.581395	0.574713

Logistic Regression Classification

In [65]:

```
# Logistic Regression Classification

lr_results = []

for combo_name, features in classification_combos.items():
    X_train_combo = X_train_scaled_df[features]
    X_test_combo = X_test_scaled_df[features]

    lr_model = LogisticRegression(max_iter=1000, random_state=42)

    lr_model.fit(X_train_combo, y_train_class)
    y_pred = lr_model.predict(X_test_combo)

    lr_results.append({
        "Model": "Logistic Regression",
        "Features": combo_name,
        "Accuracy": accuracy_score(y_test_class, y_pred),
        "Precision": precision_score(y_test_class, y_pred, zero_division=0),
        "Recall": recall_score(y_test_class, y_pred, zero_division=0),
        "F1 Score": f1_score(y_test_class, y_pred, zero_division=0)
    })

lr_results_df = pd.DataFrame(lr_results)
lr_results_df.sort_values(by="F1 Score", ascending=False)
```

Out[65]:

	Model	Features	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	All 11 features	0.89375	0.736842	0.325581	0.451613
1	Logistic Regression	Top 5 by correlation	0.89375	0.764706	0.302326	0.433333

Summary Table + Best Model + Confusion Matrix

In [66]:

```
# Combine all results
all_results = pd.concat([
    dt_results_df[['Model','Features','Accuracy','Precision','Recall','F1 Score']],
    knn_results_df[['Model','Features','Accuracy','Precision','Recall','F1 Score']],
    nb_results_df[['Model','Features','Accuracy','Precision','Recall','F1 Score']],
    lr_results_df[['Model','Features','Accuracy','Precision','Recall','F1 Score']]
], ignore_index=True)

# Show top results
summary_table = all_results.sort_values(by='F1 Score', ascending=False)
print("Classification Model Comparison:")
display(summary_table)

# Best model = Decision Tree
best_model = DecisionTreeClassifier(
    criterion='entropy',
    max_depth=None,
    min_samples_split=5,
    random_state=42
)

best_model.fit(X_train_scaled_df[feature_cols], y_train_class)
y_best_pred = best_model.predict(X_test_scaled_df[feature_cols])

cm = confusion_matrix(y_test_class, y_best_pred)

print("Confusion Matrix (Best Model: Decision Tree)")
print(cm)
print()
print(classification_report(y_test_class, y_best_pred))
```

```
Classification Model Comparison:
```

	Model	Features	Accuracy	Precision	Recall	F1 Score
9	Decision Tree	All 11 features	0.925000	0.731707	0.697674	0.714286
10	Decision Tree	All 11 features	0.925000	0.731707	0.697674	0.714286
11	Decision Tree	All 11 features	0.918750	0.717949	0.651163	0.682927
6	Decision Tree	All 11 features	0.912500	0.692308	0.627907	0.658537
7	Decision Tree	All 11 features	0.909375	0.694444	0.581395	0.632911
8	Decision Tree	All 11 features	0.906250	0.675676	0.581395	0.625000
4	Decision Tree	All 11 features	0.906250	0.709677	0.511628	0.594595
5	Decision Tree	All 11 features	0.906250	0.709677	0.511628	0.594595
32	Naive Bayes	All 11 features	0.865625	0.500000	0.720930	0.590476
3	Decision Tree	All 11 features	0.903125	0.687500	0.511628	0.586667
33	Naive Bayes	Top 5 by correlation	0.884375	0.568182	0.581395	0.574713
22	Decision Tree	Top 5 by correlation	0.884375	0.568182	0.581395	0.574713
18	Decision Tree	Top 5 by correlation	0.884375	0.571429	0.558140	0.564706
20	Decision Tree	Top 5 by correlation	0.890625	0.611111	0.511628	0.556962
21	Decision Tree	Top 5 by correlation	0.884375	0.575000	0.534884	0.554217
17	Decision Tree	Top 5 by correlation	0.903125	0.730769	0.441860	0.550725
16	Decision Tree	Top 5 by correlation	0.903125	0.730769	0.441860	0.550725
19	Decision Tree	Top 5 by correlation	0.881250	0.560976	0.534884	0.547619
15	Decision Tree	Top 5 by correlation	0.900000	0.703704	0.441860	0.542857
23	Decision Tree	Top 5 by correlation	0.887500	0.600000	0.488372	0.538462
2	Decision Tree	All 11 features	0.893750	0.714286	0.348837	0.468750
1	Decision Tree	All 11 features	0.893750	0.714286	0.348837	0.468750
0	Decision Tree	All 11 features	0.893750	0.714286	0.348837	0.468750
14	Decision Tree	Top 5 by correlation	0.893750	0.714286	0.348837	0.468750
13	Decision Tree	Top 5 by correlation	0.893750	0.714286	0.348837	0.468750
12	Decision Tree	Top 5 by correlation	0.893750	0.714286	0.348837	0.468750
28	KNN	Top 5 by correlation	0.871875	0.531250	0.395349	0.453333
34	Logistic Regression	All 11 features	0.893750	0.736842	0.325581	0.451613
35	Logistic Regression	Top 5 by correlation	0.893750	0.764706	0.302326	0.433333
29	KNN	Top 5 by correlation	0.865625	0.500000	0.325581	0.394366
24	KNN	All 11 features	0.840625	0.388889	0.325581	0.354430
26	KNN	All 11 features	0.871875	0.550000	0.255814	0.349206
30	KNN	Top 5 by correlation	0.868750	0.526316	0.232558	0.322581
25	KNN	All 11 features	0.859375	0.454545	0.232558	0.307692
31	KNN	Top 5 by correlation	0.871875	0.600000	0.139535	0.226415
27	KNN	All 11 features	0.859375	0.400000	0.093023	0.150943

```
Confusion Matrix (Best Model: Decision Tree)
[[266  11]
 [ 13  30]]

              precision    recall  f1-score   support

           0       0.95      0.96      0.96       277
           1       0.73      0.70      0.71        43

    accuracy                           0.93       320
   macro avg       0.84      0.83      0.84       320
weighted avg       0.92      0.93      0.92       320
```

Summary of Classification Results

The classification task compared four models: Decision Tree, KNN, Naive Bayes and and Logistic Regression using two feature sets (all 11 features and top 5 correlated features).

The best performing model was the Decision Tree classifier using all 11 features with max_depth=None and min_samples_split=5. It achieved an Accuracy of 0.9250, Precision of 0.7317, Recall of 0.6977, and the highest F1 Score of 0.7143. == 92.5% accuracy, 73.2% precision, 69.8% recall, and 71.4% F1 score.

Naive Bayes performed moderately well, reaching an F1 score of 59.0%, while KNN had the weakest performance overall.

The KNN model performed worse than Decision Tree. Its best result came from using the Top 5 correlated features with k=3, producing an Accuracy of 0.8719 and F1 Score of 0.4533.

Logistic Regression showed similar performance to KNN but lower than Decision Tree. Its best model used all 11 features and achieved an Accuracy of 0.8938 with an F1 Score of 0.4516.

Overall, the Decision Tree model was the strongest classifier because it provided the best balance between precision and recall, resulting in the highest F1 Score.

Conclusion | Summary: (View Report)